BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 086 · Lesson 086 · Debugging Unfamiliar Algorithms

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is a loop invariant?

Q2. What is a boundary test?

Q3. What return value commonly indicates a failed indexed search?

Q4. In array [2,4,6], set a[1]=a[0]+a[2]. Give the final array.

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I Do · Read new knowledge

Key vocabulary

logic error — Program runs but produces wrong answer.

boundary — Edge of valid range.

trace — Record changing states.

debugging — Finding and correcting faults.

New learning

Observe — Trace before changing code: list variables, loop indexes, condition and expected return.

Explain — An off-by-one fault often comes from <= Length instead of < Length; an empty array must also be considered.

Evaluate — Correct code by explaining the cause and showing a fresh test, not just copying a working version.

Worked example

Repair for-loop condition i<=a.Length in an array scan.

1. Model response — For length4, valid indexes0–3. Checking a[4] would throw.

We Do · Practise with guidance

A separate function starts count=1 when counting even numbers in [1,3,5]. Fix.

Model response

Initialising to1 incorrectly counts an item before any check.

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Trace CountEven([2,5,8,9]) with count after each item. [6]

Question 2

Explain why i<=a.Length is unsafe in a loop indexing a[i], and repair. [6]

Question 3

Develop four distinct tests for a function counting even values, with expected answers and reasons. [8]

Exit Ticket · Answer before checking

A loop using i<=Length can safely read a[i] for every i.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Property maintained after each iteration.

Q2. A test at or next to a threshold.

Q3. −1.

Q4. [2,8,6].

You Do pages · Independent questions

Q1. After2→1 [2],5→1 [1],8→2 [1],9→2 [1], return2 [1].

Q2. Last index Length−1 [2], i=Length is invalid [2], use i<Length [2].

Q3. Empty[]→0 [2], all odd[1,3]→0 [2], all even[2,4]→2 [2], mixed [1,2,3]→1 [2].

Exit Ticket · Model answer

False. The final index is out of bounds.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: